Sequences & Patterns

ANSWER KEY

Section 1: Arithmetic Sequences & Constant Differences

1 What is the next number in the arithmetic sequence: $4, 11, 18, 25, \ldots$?

a. 32 ✓

b. 31

c. 33

d. 35

Common difference $d = 11 - 4 = 7$. Next term $= 25 + 7 = 32$.

2 In an arithmetic sequence where the first term is 3 and the common difference is 4, what is the 15th term?

a. 59 ✓

b. 63

c. 55

d. 60

$a_n = a_1 + (n-1)d = 3 + (15-1)(4) = 3 + 14(4) = 3 + 56 = 59$.

3 What is the sum of the first 20 positive even integers $(2 + 4 + 6 + \ldots + 40)$?

a. 420 ✓

b. 400

c. 440

d. 380

Formula for the sum of the first n even integers: $S_n = n(n+1)$. Here $n = 20$, so $S_{20} = 20 \times 21 = 420$.

Section 2: Geometric & Exponential Growth Sequences

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4 What is the next term in the geometric sequence: $3, 6, 12, 24, \ldots$?

a. 48 ✓

b. 36

c. 42

d. 54

Common ratio $r = \frac{6}{3} = 2$. Next term $= 24 \times 2 = 48$.

5 A bacterial culture doubles in number every 20 minutes. If there are initially 100 bacteria, how many will there be after 2 hours?

a. 6400 ✓

b. 3200

c. 12800

d. 1600

2 hours $= 120$ minutes. Number of doubling cycles $= \frac{120}{20} = 6$. Population $= 100 \times 2^6 = 100 \times 64 = 6400$.

6 What is the 6th term of the sequence: $2, -6, 18, -54, \ldots$?

a. -486 ✓

b. 486

c. -162

d. 162

Common ratio $r = -3$. $a_6 = a_1 r^5 = 2(-3)^5 = 2(-243) = -486$.

Section 3: Composite, Alternating & Two-Step Patterns

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7 Find the missing number in the sequence: $2, 3, 5, 8, 13, 21, \ldots$?

a. 34 ✓

b. 32

c. 33

d. 35

This is a Fibonacci pattern where each term is the sum of the two preceding terms: $13 + 21 = 34$.

8 What is the next number in the pattern: $1, 4, 9, 16, 25, \ldots$?

a. 36 ✓

b. 35

c. 40

d. 49

These are consecutive perfect squares: $1^2, 2^2, 3^2, 4^2, 5^2, 6^2 = 36$.

9 Find the next term in the sequence: $2, 5, 11, 23, 47, \ldots$?

a. 95 ✓

b. 94

c. 91

d. 89

Each term is generated by multiplying the preceding term by 2 and adding 1: $47 \times 2 + 1 = 94 + 1 = 95$.

Section 4: Periodic Cycles & Calendar Patterns

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10 If today is Tuesday, what day of the week will it be 100 days from now?

a. **Thursday ✓**

b. Wednesday

c. Friday

d. Tuesday

Days of the week repeat every 7 days. Divide $100 \div 7 = 14$ with remainder 2. Two days after Tuesday is Thursday.

11 A light flashes every 12 seconds, and a second light flashes every 18 seconds. If they flash together right now, after how many seconds will they flash together again?

a. 36 **seconds ✓**

b. 24 seconds

c. 48 seconds

d. 72 seconds

The lights flash together at multiples of the least common multiple: $\text{LCM}(12, 18) = 36$ seconds.

12 In a repeating decimal $0.\overline{142857}$, what is the 50th digit after the decimal point?

a. 8 ✓

b. 4

c. 2

d. 5

The repeating block has length 6 $(1, 4, 2, 8, 5, 7)$. Dividing $50 \div 6 = 8$ with remainder 2. The 2nd digit in the cycle is 4... wait, the 4th is 8! $50 = 6 \times 8 + 2$. The 2nd digit is 4.

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